higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION

(First Paper)

NQF LEVEL 2

(6031002)

12 March 2019 (X-Paper)
09:00–11:00

Calculators and drawing instruments may be used.

This question paper consists of 5 pages.

NCV

TIME: 2 HOURS
MARKS: 75

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Clearly cross out ALL work you do NOT want to be marked.
5. Write neatly and legibly.

NC(V)

QUESTION 1

1.1 Templates provide shapes which might be marked off to produce duplicates of patterns with continuous repetition.

Differentiate between a *template* and a *pattern*. (2 + 2) (4)

1.2 Indicate whether the following statement is TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.2) in the ANSWER BOOK.

Using a template as a guide plate for drilling into steel sections is one of the reasons for templating. (1)

1.3 Illustrate with a neat sketch how a template is used to mark off holes on a channel. (4)

1.4 List THREE developmental methods boilermakers use when making patterns. (3)

1.5 Illustrate with a neat sketch a typical example of each developmental method mentioned in QUESTION 1.4. (3)

[15]

QUESTION 2

2.1 During a pipe-rolling process the mean line plays an important role.

Differentiate between the *mean diameter* and the *inside diameter* formed during the rolling process. (2 + 2) (4)

2.2 Metal properties play an important role during a pipe-rolling process.

State TWO functions of malleability as a metal property. (2)

2.3 A 10 mm mild-steel plate is used to roll up a cylinder with an inside diameter of 1 000 mm and a height of 1 800 mm.

2.3.1 Illustrate with a neat sketch the cylinder in QUESTION 2.3. (4)

2.3.2 Calculate the mean diameter of the cylinder in QUESTION 2.3. (1)

2.3.3 Calculate the area of the plate required to up the cylinder in QUESTION 2.3. (3)

2.4 Indicate whether the following statement is TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.4) in the ANSWER BOOK.

Mean diameter refers to a plate thickness added to an outside diameter. (1)

[15]

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QUESTION 3

3.1 Boilermakers use bending and forming machines during the fabrication process.

3.1.1 List THREE types of bending machines. (3)

3.1.2 Differentiate between a *bending machine* and a *forming (rolling) machine*. (2 + 2) (4)

3.2 Illustrate with a sketch how a mild-steel plate is rolled in a forming machine. (6)

3.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (3.3.1–3.3 2) in the ANSWER BOOK.

3.3.1 The rolling process takes time until the pipe takes up the required shape.

3.3.2 There is no difference between a press brake and a bar folder.

(2 × 1) (2)

[15]

QUESTION 4

4.1 Oxy-acetylene is extensively used mostly for cutting metal to required shapes and sizes.

4.1.1 Name TWO cylinders involved during in an oxy-acetylene cutting process. (2)

4.1.2 Compare the TWO cylinders mentioned in QUESTION 4.1.1 by completing the table below. Write only the answer next to the letter (A–D) in the ANSWER BOOK.

ITEM	CYLINDER 1	CYLINDER 2
Cylinder name	A	B
Colour	C	Maroon
Thread direction	Right-handed	D

(4 × 1) (4)

4.2 A neutral flame is one of the flames formed during an oxy-acetylene flame setting.

Illustrate with neat sketches the difference between *carburising* and *oxidising* flames. (2 + 2) (4)

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4.3 Indicate whether the following statement is TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (4.3) in the ANSWER BOOK.

Oxidising flames are mostly used for welding most metals. (1)

4.4 Illustrate with a neat sketch how each of the following surfaces will appear if:

4.4.1 The nozzle is held too high from the metal being cut.

4.4.2 The nozzle is held too close to the metal being cut.

(2 × 2) (4)
[15]

QUESTION 5

5.1 Give FIVE advantages of using a coolant. (5)

5.2 Compare an *abrasive cutting saw* to a *friction saw*. (2 + 2) (4)

5.3 Illustrate with a neat sketch how each of the following steel sections are wrongly held when cutting a disc:

5.3.1 T-bar

5.3.2 Flat bar

5.3.3 Angle iron

(3 × 2) (6)
[15]

TOTAL: 75

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